

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 11

Assessment 2 Term 2 2024

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators may be used
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination.

Total marks – 45

Section I Pages 2 – 3

5 marks

Attempt Questions 1 – 5

Answer on the Objective Response Answer Sheet provided

Section II Pages 4 – 7

40 marks

Attempt Questions 6 – 9.

Question	1-5	6	7	8	9	Total
Total	/5	/10	/10	/10	/10	/45

This assessment task constitutes 35% of the Preliminary Higher School Certificate Course School Assessment.

Outcomes Examined ME11-1, ME11-2, ME11-3, MA11-1, MA11-2, MA11-3, MA11-4, MA11-9, ME11-7

Section I

5 marks

Attempt Questions 1 – 5

Allow about 10 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

Q1. If $\tan \theta = \frac{3}{4}$ and $\cos \theta < 0$ then $\sin \theta$ is:

(A) $\frac{3}{5}$

(B) $\frac{4}{5}$

(C) $-\frac{3}{5}$

(D) $-\frac{4}{5}$

Q2. The solution of $x^2 - 3x < 4$ is:

(A) $\{x: 1 < x < -4\}$

(B) $\{x: -1 < x < 4\}$

(C) $\{x: x < -1, x > 4\}$

(D) $\{x: x < -4, x > 1\}$

Q3. $\frac{1 - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$ is equivalent to:

(A) 1

(B) $\sec^2 \theta + \cos^2 \theta$

(C) $\tan^2 \theta + \sin^2 \theta$

(D) $\tan^2 \theta + \cos^2 \theta$

Q4. Which of the following pairs of functions are NOT inverse functions?

(A) $f(x) = 2x - 1$ and $g(x) = \frac{1}{2}(x + 1)$

(B) $f(x) = \frac{1}{2x-1}, x > \frac{1}{2}$ and $g(x) = \frac{2x+1}{x}, x > 0$

(C) $f(x) = x^2 - 1, x \geq 0$ and $g(x) = \sqrt{x+1}, x \geq -1$

(D) $f(x) = x^2 - 1, x \leq 0$ and $g(x) = -\sqrt{x+1}, x \geq -1$

Q5. The Parametric equations $x = 6 \cos \theta$ and $y = 6 \sin \theta$ when written in Cartesian form as:

(A) $x^2 + y^2 = 6$

(B) $x^2 - y^2 = 6$

(C) $x^2 - y^2 = 36$

(D) $x^2 + y^2 = 36$

End of Section I

Section II

40 marks

Attempt Questions 6-8

Allow about 50 minutes for this section

Question 6 (10 marks)

Start a new answer booklet

(a) Showing full working, rationalise the denominator of $\frac{3+2\sqrt{5}}{2\sqrt{5}-1}$ 2

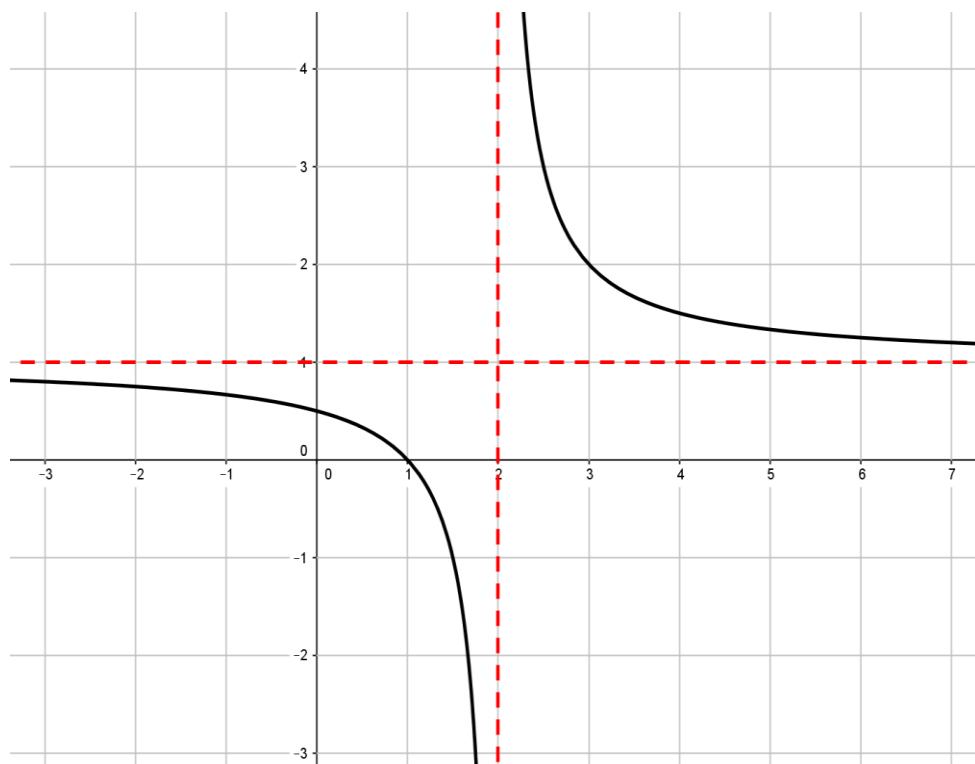
(b) Simplify $\frac{\sin \theta}{1+\cos \theta} + \frac{1+\cos \theta}{\sin \theta}$ 2

(c) Solve $|x^2 - 1| \leq 4$ 3

(d) Simplify $\frac{x^3-1}{x^2-1} \div \frac{3x^2+3x+3}{x^2-4x-5}$ 3

Question 7 (10 marks)**Start a new answer booklet**

- (a) The diagram below is a sketch of the graph $y = f(x)$.



Use the graphs provided on the separate answer sheet, to graph the following transformations showing all asymptotes and intercepts.

(i) $y = |f(x)|$ **1**

(ii) $y^2 = f(x)$ **2**

(iii) $y = \frac{1}{f(x)}$ **2**

(iv) If $g(x) = 2x$, sketch $y = g[f(x)]$ **2**

(b) Solve $\cos 2\theta = \frac{1}{2}$ for $0 \leq \theta \leq 2\pi$. **3**

Question 8 (10 marks)**Start a new answer booklet**

- (a) Find the value of $\frac{a}{b^3c^2}$ in index form if: 2

$$a = \left(\frac{2}{5}\right)^4, \quad b = \left(-\frac{1}{3}\right)^3 \quad \text{and} \quad c = \left(\frac{3}{5}\right)^2$$

- (b) Prove that $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$ 2

- (c) (i) Find the domain and range of the function $f(x) = \sqrt{5-x} - 1$. 2

- (ii) Find the equation of its inverse $f^{-1}(x)$. 2

- (iii) Find the intersecting point of $f(x)$ and $f^{-1}(x)$. 2

Question 9 (10 marks)**Start a new answer booklet**

(a) Solve $5\cos^2 x + 2\sin x = 2$ for $0^\circ \leq x \leq 180^\circ$ **3**

(b) Find the cartesian equation of $x = 2p + 3, y = p^2$ and describe any important features of the equation **2**

(c) Write $x^2 + y^2 - 6x + 2y - 26 = 0$ in parametric form with parameter θ . **2**

(d) Find the values of x for which the inequalities are simultaneously satisfied: **3**

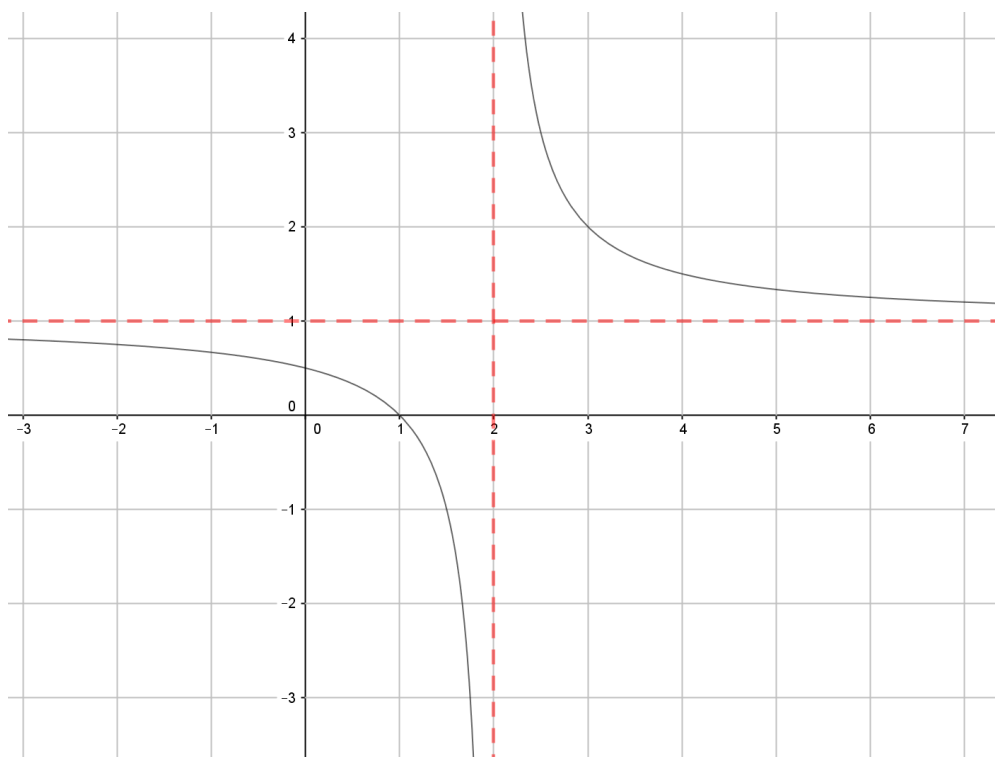
$$\frac{7}{(3-x)(x+3)} > -1 \quad \text{and} \quad x^2 + 3x - 10 < 0$$

End of Assessment

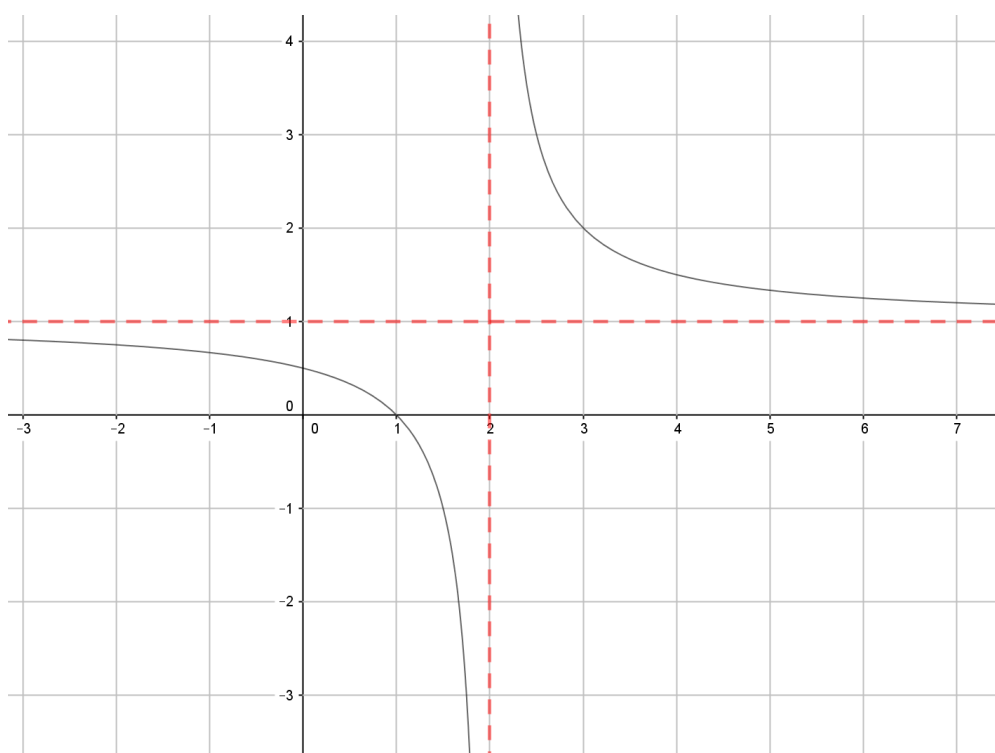
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Question 7(a) Place this sheet inside the Question 7 Answer Booklet

(i) $y = |f(x)|$

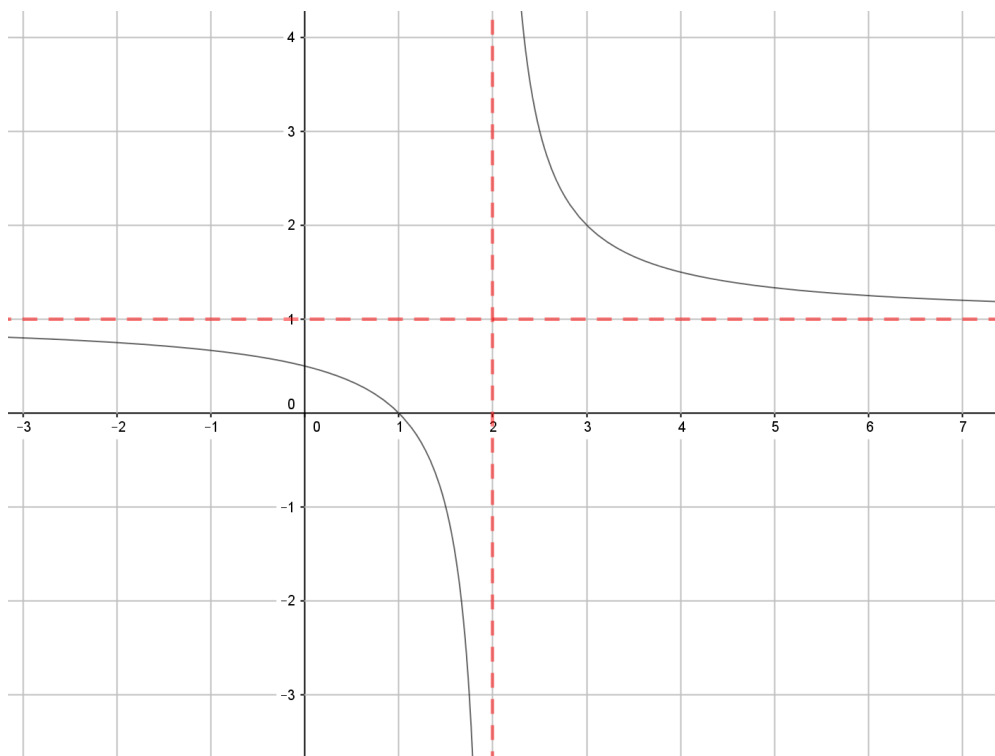


(ii) $y^2 = f(x)$

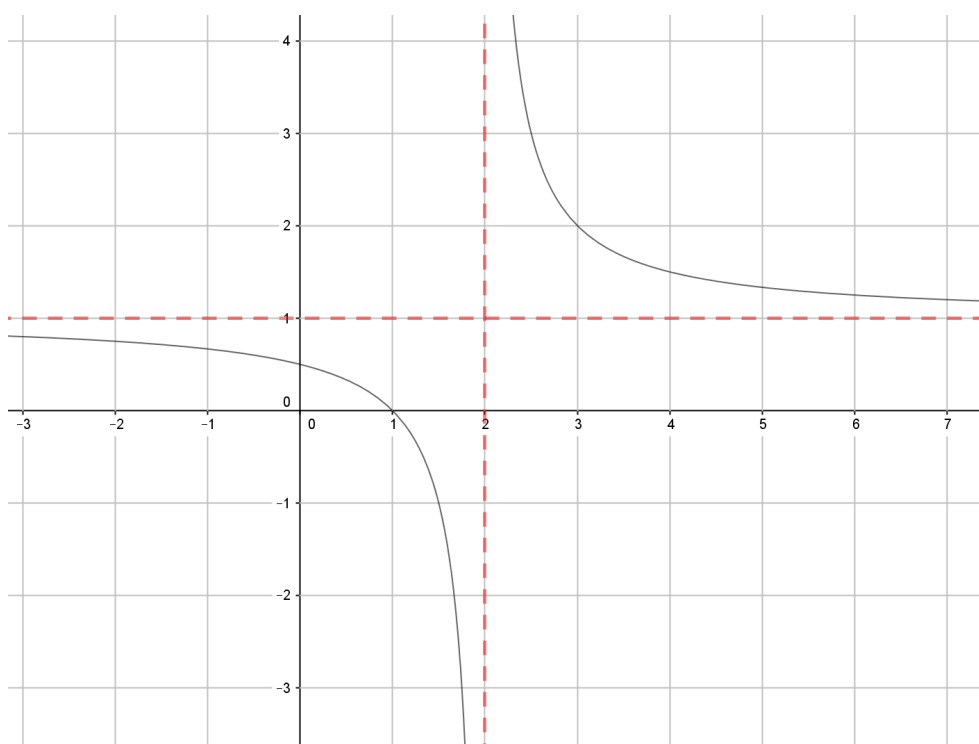


Question 7(a)

(iii) $y = \frac{1}{f(x)}$



(iv) If $g(x) = 2x$, sketch $y = g[f(x)]$

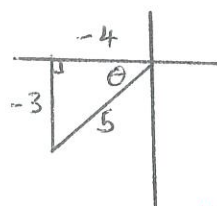


YEAR 11 Ext 1 Task 2 Solutions

and Feedback

Q1. $\tan \frac{3}{4} > 0$ and $\cos \theta < 0 \therefore$ third quad

$$\sin \theta = -\frac{3}{5}$$



(C)

Mostly well done.

*. Incorrect choice of A when forgotten it is in QIII

(B)

Mostly well done.

Incorrect choice of (C) student did not use a graph or number line to obtain answer.

(D)

50% obtained the correct answer.

25% chose incorrectly the answer (C)

Q2. $x^2 - 3x < 4$
 $x^2 - 3x - 4 < 0$
 $(x-4)(x+1) < 0$



$$\{x: -1 < x < 4\}$$

Q3. $\frac{1 - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$

$$= \sec^2 \theta - \sin^2 \theta \quad \text{Not an option}$$

$$= 1 + \tan^2 \theta - \sin^2 \theta \quad \text{" " "}$$

$$= \tan^2 \theta + \cos^2 \theta \quad \checkmark$$

Q4. (A) $f(x) = 2x - 1$
 $y = 2x - 1$

$\therefore$ inv $x = 2y - 1$

$$2y = x + 1$$

$$y = \frac{1}{2}(x+1)$$

$$g(x) = \frac{1}{2}(x+1) \quad \checkmark$$

$\therefore$ they are a pair of inverse fns.

(B) $f(x) = \frac{1}{2x-1}, x > \frac{1}{2} \quad f(x) > 0$

$$y = \frac{1}{2x-1}$$

$\therefore$ inv $x = \frac{1}{2y-1}$

$$2y - 1 = \frac{1}{x}$$

$$2y = \frac{1}{x} + 1$$

$$y = \frac{1}{2x} + \frac{1}{2}$$

$$= \frac{1+x}{2x}$$

$$\neq \frac{2x+1}{x} \therefore \text{Not an inverse fn.}$$

(B)

66% obtained correct answer.

approx. 25% obtained incorrectly (D)

No need to look at (C) & (D) but

the working is over the page to confirm they are pairs of inverse functions.

Q5. $x = 6 \cos \theta \Rightarrow \cos \theta = \frac{x}{6} \quad y = 6 \sin \theta \Rightarrow \sin \theta = \frac{y}{6}$

using $\cos^2 \theta + \sin^2 \theta = 1$

$$\frac{x^2}{36} + \frac{y^2}{36} = 1$$

$$x^2 + y^2 = 36$$

$\therefore$ (D)

Mostly done well.

Q4 (c) $f(x) = x^2 - 1$, $x \geq 0$, $y \geq -1$
 $y = x^2 - 1$

$\therefore$ inv: $x = y^2 - 1$

$y^2 = x + 1$

$y = \pm \sqrt{x+1}$ but $x \geq -1$ & $y \geq 0$

$\therefore y = \sqrt{x+1}$

$\therefore g(x) = \sqrt{x+1}$

$\therefore f(x)$ & $g(x)$ are inverse fns

(D) $f(x) = x^2 - 1$, $x \leq 0$, $y \geq -1$
 $y = x^2 - 1$

$\therefore$ inv: $x = y^2 - 1$

$y^2 = x + 1$

$y = \pm \sqrt{x+1}$ but $x \geq -1$ & $y \leq 0$

$\therefore y = -\sqrt{x+1}$

$\therefore g(x) = -\sqrt{x+1}$

$\therefore f(x)$ & $g(x)$ are inverse fns.

* For Q4 you could also show that

$f(g(x)) = g(f(x)) = x$ for (A), (C) & (D)

but not for (B)

Question 6

$$(a) \frac{(3+2\sqrt{5})}{(2\sqrt{5}-1)} \times \frac{(2\sqrt{5}+1)}{(2\sqrt{5}+1)} = \frac{6\sqrt{5}+3+20+2\sqrt{5}}{20-1}$$

$$= \frac{8\sqrt{5}+23}{19}$$

Mostly done well.
Just reminding everyone
conjugate of $(2\sqrt{5}-1)$ is
 $(2\sqrt{5}+1)$, not $(\sqrt{5}+1)$

$$(b) \frac{\sin \theta}{1+\cos \theta} + \frac{1+\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + (1+\cos \theta)^2}{\sin \theta (1+\cos \theta)}$$

$$= \frac{\sin^2 \theta + 1 + 2\cos \theta + \cos^2 \theta}{\sin \theta (1+\cos \theta)}$$

$$= \frac{2 + 2\cos \theta}{\sin \theta (1+\cos \theta)} \quad (\sin^2 \theta + \cos^2 \theta = 1)$$

$$= \frac{2(1+\cos \theta)}{\sin \theta (1+\cos \theta)}$$

$$= \frac{2}{\sin \theta}$$

$$= 2 \operatorname{cosec} \theta$$

Mostly done well.

$$(c) |x^2-1| \leq 4$$

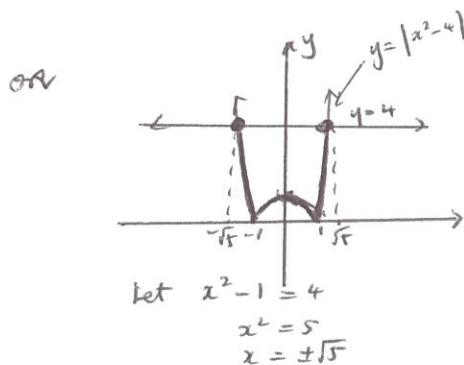
$$-4 \leq x^2-1 \leq 4$$

$$-4 \leq x^2-1 \quad \text{and} \quad x^2-1 \leq 4$$

$$-3 \leq x^2 \quad x^2 \leq 5$$

$$\text{all } x \quad \text{and} \quad -\sqrt{5} \leq x \leq \sqrt{5}$$

$$\therefore \{x: -\sqrt{5} \leq x \leq \sqrt{5}\}$$



∴ As graphically shown,
 $\{x: -\sqrt{5} \leq x \leq \sqrt{5}\}$

Mostly done well.
If a graph is used as
a tool, then expected
to sketch $y=|x^2-4|$

$$(d) \frac{x^3-1}{x^2-1} \div \frac{3x^2+3x+3}{x^2-4x-5} = \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} \times \frac{(x-5)(x+1)}{3(x^2+x+1)}$$

$$= \frac{x-5}{3}$$

Mostly done well.

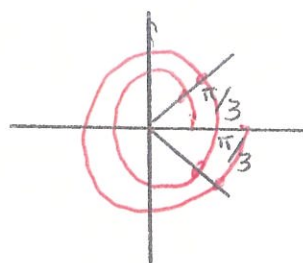
Question 7

(a) (i) - (iv) See Separate Sheet.

$$(b) \quad \cos 2\theta = \frac{1}{2} \quad \begin{aligned} 0^\circ \leq \theta \leq 2\pi^\circ \\ 0^\circ \leq 2\theta \leq 4\pi^\circ \end{aligned}$$

$$2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$



Generally well done.
Some students didn't adjust the domain for 2θ .

Question 8

See Next page for markers comments + extra solutions.

$$(a) \quad \frac{a}{b^3 c^2} = \frac{\left(\frac{2}{5}\right)^4}{\left[\left(-\frac{1}{3}\right)^3\right]^3 \left[\left(\frac{3}{5}\right)^2\right]^2}$$

$$= \frac{2^4}{5^4}$$

$$\frac{-1}{3^{27}} \cdot \frac{3^4}{5^4}$$

$$= \frac{2^4}{5^4} \cdot \frac{3^5 \cdot 5^4}{-1^9}$$

$$= -2^4 \cdot 3^5$$

RTP

$$(b) \quad \sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$$

$$\text{LHS} = \sec \theta + \tan \theta$$

$$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \sin \theta}{\cos \theta}$$

$$= \text{RHS}$$

$$(c) \quad (i) \quad f(x) = \sqrt{5-x} - 1$$

$$D: 5-x \geq 0$$

$$-x \geq -5$$

$$x \leq 5$$

$$R: y \geq -1$$

$$(ii) \quad \text{let } y = f(x)$$

$$\therefore y = \sqrt{5-x} - 1$$

$$\therefore \text{inv } x = \sqrt{5-y} - 1$$

$$x+1 = \sqrt{5-y}$$

$$(x+1)^2 = 5-y$$

$$y = 5 - (x+1)^2 \quad D: x \geq -1$$

$$f^{-1}(x) = 5 - (x+1)^2 \quad R: y \leq 5$$

(iii) $f(x)$ & $f^{-1}(x)$ will intersect on $y=x$ if they do intersect.

$$\therefore \sqrt{5-x} - 1 = x$$

$$\sqrt{5-x} = x+1$$

$$5-x = x^2 + 2x + 1$$

$$0 = x^2 + 3x - 4$$

$$0 = (x+4)(x-1)$$

$\therefore x = -4$ & $x = 1$
but only if $-1 \leq x \leq 5$ (see Domains)
 $\therefore$ when $x=1, y=1$ (1,1)

Question 8

a) if $a = \left(\frac{2}{5}\right)^4$, $b = \left(-\frac{1}{3}\right)^3$, $c = \left(\frac{3}{5}\right)^2$

Common error

then $\frac{a}{b^3 c^2} = \frac{\left(\frac{2}{5}\right)^4}{\left(\left(-\frac{1}{3}\right)^3\right)^3 \left(\left(\frac{3}{5}\right)^2\right)^2}$

$= \frac{\left(\frac{2}{5}\right)^4}{\left(-\frac{1}{3}\right)^9 \left(\frac{3}{5}\right)^4}$

$= \frac{2^4}{5^4} \div 5^4$

$= \frac{(-1)^9 \times \frac{3^4}{5^4}}{3^9 \times \frac{3^4}{5^4}} \div 5^4$

$= \frac{2^4}{\frac{-1}{3^9} \times 3^4}$

$= \frac{2^4}{-1 \times 3^{-5}}$

$= -2^4 \times 3^5$

must be in index form
(= -3888)

1 for correct substitution in index form and some correct manipulation in index form

NOT

$\frac{a}{bc}$

$= \frac{\left(\frac{2}{5}\right)^4}{\left(-\frac{1}{3}\right)^3 \left(\frac{3}{5}\right)^2}$

$= \frac{2^4}{5^4} \div \left(\frac{(-1)^3 \times 3^2}{3^3 \times 5^2}\right)$

$= \frac{2^4}{5^4} \times \frac{3^3 \times 5^2}{-1 \times 3^2}$

$= \frac{2^4 \times 3 \times 5^2}{-5^4}$

$= -\frac{2^4 \times 3}{5^2}$

$= \frac{48}{-25}$ 1 mark only

b) Prove that $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$

LHS = $\sec \theta + \tan \theta$

← Start here, rewriting LHS of Q or RHS

$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$

← NOT here

$= \frac{1 + \sin \theta}{\cos \theta}$

1 for interpreting $\sec \theta$ and $\tan \theta$
1 for expressing as a single fraction

LHS = RHS

RHS = $\frac{1 + \sin \theta}{\cos \theta}$

$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$ 1 for separating fraction

$= \sec \theta + \tan \theta$

RHS = LHS

8 c) i) $f(x) = \sqrt{5-x} - 1$

D: $5-x \geq 0$
 $-x \geq -5$
 $x \leq 5$ or $(-\infty, 5]$

R: $y \geq -1$ or $[-1, \infty)$

ii) $f: y = \sqrt{5-x} - 1$

$f^{-1}: x = \sqrt{5-y} - 1$

$x+1 = \sqrt{5-y}$

$(x+1)^2 = 5-y$

$y = 5 - (x+1)^2$

$= 5 - (x^2 + 2x + 1)$

$y = -x^2 - 2x + 4$

$f^{-1}(x) = 5 - (x+1)^2$

NOT $x^2 = 5-y-1$ $\times$
or $5-y+1$

2 marks awarded for $y = 5 - (x+1)^2$

very few students correctly stated

$f^{-1}(x) = 5 - (x+1)^2$
 $= -x^2 - 2x + 4$

D: $x \geq -1$

R: $y \leq 5$

iii) A function and its inverse intersect on $y=x$ so solve

$\sqrt{5-x} - 1 = x$

$\sqrt{5-x} = x+1$

$5-x = (x+1)^2$

$5-x = x^2 + 2x + 1$

$0 = x^2 + 3x - 4$

$0 = (x+4)(x-1)$

$x = -4, 1$

or

$5 - (x+1)^2 = x$

$5 - (x^2 + 2x + 1) = x$

$5 - x^2 - 2x - 1 = x$

$-x^2 - 3x + 4 = 0$

$x^2 + 3x - 4 = 0$

$(x+4)(x-1) = 0$

$x = -4, 1$

1 for finding $x = -4, 1$

but D for $f^{-1}(x)$ is $x \geq -1$
 $\therefore x \neq -4$

but D for $f^{-1}(x)$ is $x \geq -1$

$\therefore x \neq -4$

The only point of intersection is $(1, 1)$

1 for giving reasons to discard $x = -4$.

The only point of intersection is $(1, 1)$

Solving $f(x) = f^{-1}(x)$

$\sqrt{5-x} - 1 = 5 - (x+1)^2$

is inefficient and requires factorising involved polynomials.

Only one student was successful by this method.

Question 9

(a) $5\cos^2 x + 2\sin x = 2$

$0^\circ \leq x \leq 180^\circ$

$5(1 - \sin^2 x) + 2\sin x - 2 = 0$

first mark

$5 - 5\sin^2 x + 2\sin x - 2 = 0$

$5\sin^2 x - 2\sin x - 3 = 0$

$(5\sin x + 3)(\sin x - 1) = 0$

2nd mark

$\therefore \sin x = -\frac{3}{5} \text{ or } \sin x = 1$

No solutions
in domain

$x = 90^\circ$

$x = 90^\circ$

3rd mark

Generally not too bad

Some issues with
factorising correctly
(i.e. the signs)

A lot gave more
solutions than the
correct because
they didn't take
into account the
domain $0^\circ \leq x \leq 180^\circ$

(b) $x = 2p + 3 \Rightarrow 2p = x - 3$

$p = \frac{x-3}{2}$

$y = p^2$

$y = \left(\frac{x-3}{2}\right)^2$

$y = \frac{1}{4}(x-3)^2$

which is a parabola with vertex $(3, 0)$ with
vertical dilation of $\frac{1}{4}$.

Most got the equation
correct.

A lot didn't describe it
as a parabola with at least
one important feature

(c) $x^2 + y^2 - 6x + 2y - 26 = 0$

$x^2 - 6x + 9 + y^2 + 2y + 1 = 26 + 10$

$(x-3)^2 + (y+1)^2 = 36$

first mark

Most were able to get to
here other than a few
arithmetic errors.

using $\cos^2 \theta + \sin^2 \theta = 1$

and $\left(\frac{x-3}{6}\right)^2 + \left(\frac{y+1}{6}\right)^2 = 1$

* $\therefore \frac{x-3}{6} = \cos \theta$

& $\frac{y+1}{6} = \sin \theta$

second mark

$x-3 = 6\cos \theta$

$y+1 = 6\sin \theta$

$x = 6\cos \theta + 3$

$y = 6\sin \theta - 1$

These should be
or
These final answer

OR

$x = 6\sin \theta + 3$

$y = 6\cos \theta - 1$

if you set up *
other way for $\sin \theta$ & $\cos \theta$

$$(d) \frac{7}{(3-x)(x+3)} > -1$$

and

$$x^2 + 3x - 10 < 0$$

$$(x+5)(x-2) < 0$$

$$7(3-x)(x+3) > -(3-x)^2(x+3)^2$$

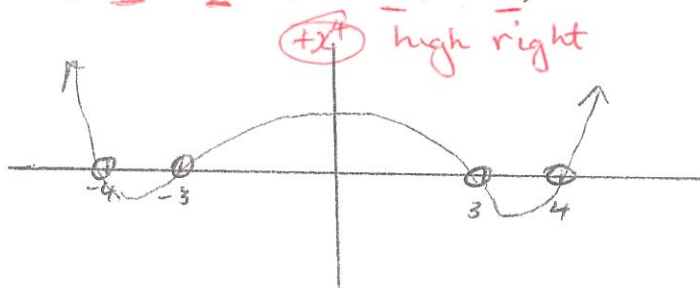
$$7(3-x)(x+3) + (3-x)^2(x+3)^2 > 0$$

$$(3-x)(x+3) [7 + (3-x)(x+3)] > 0$$

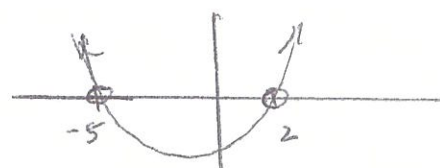
$$(3-x)(x+3) [7 + 9 - x^2] > 0$$

$$(3-x)(x+3)(16-x^2) > 0$$

$$(3-x)(x+3)(4-x)(4+x) > 0 \quad - \quad \underline{1 \text{ mark}}$$



$$x: x < -4, -3 < x < 3, x > 4$$



$$\{x: -5 < x < 2\} \quad - \quad \underline{1 \text{ mark}}$$

This one was generally well done

This uses a mixed bag. Some poor attempts, some excellent. common errors where mixing up the signs of the factors, leaving out the + underlined above. Graphing incorrectly (i.e. upside down)

Now look for intersection (and $\cap$) of both solutions.



$$\{x: -5 < x < -4, -3 < x < 2\}$$

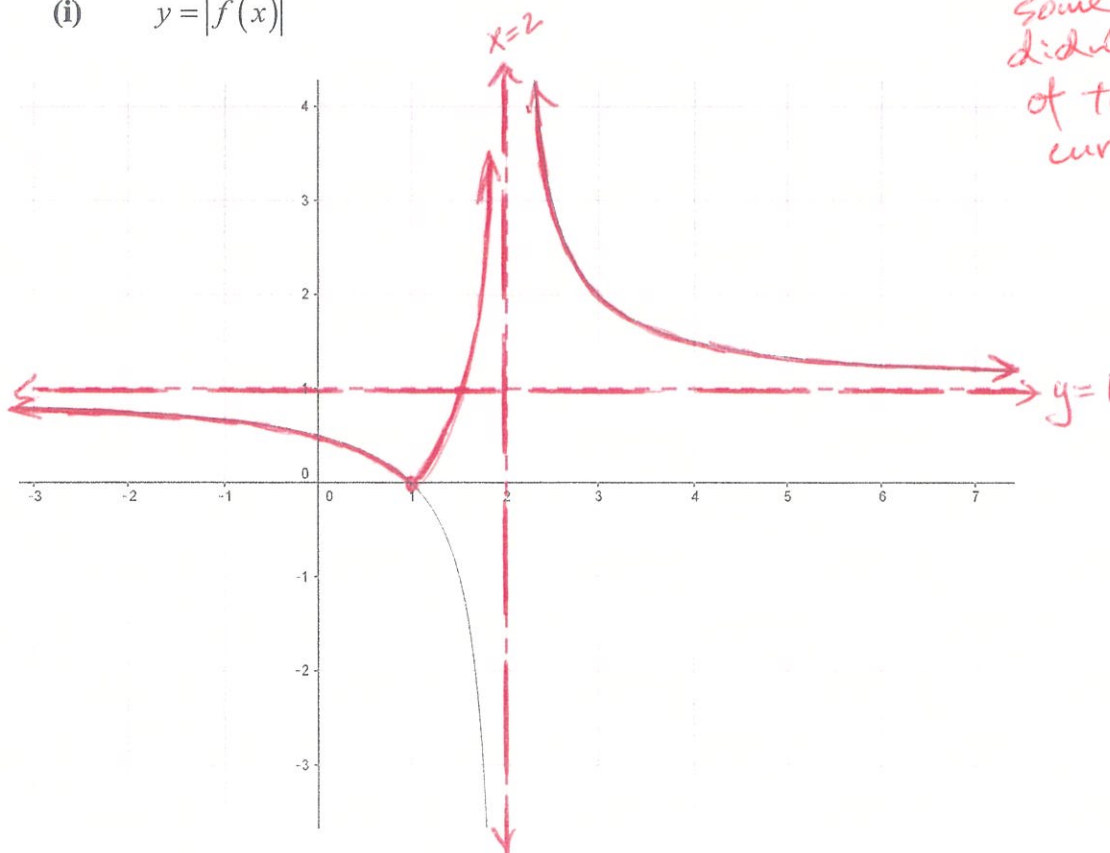
$$\text{OR } (-5, -4) \cup (-3, 2)$$

1 mark. Correct

from previous error was taken into account

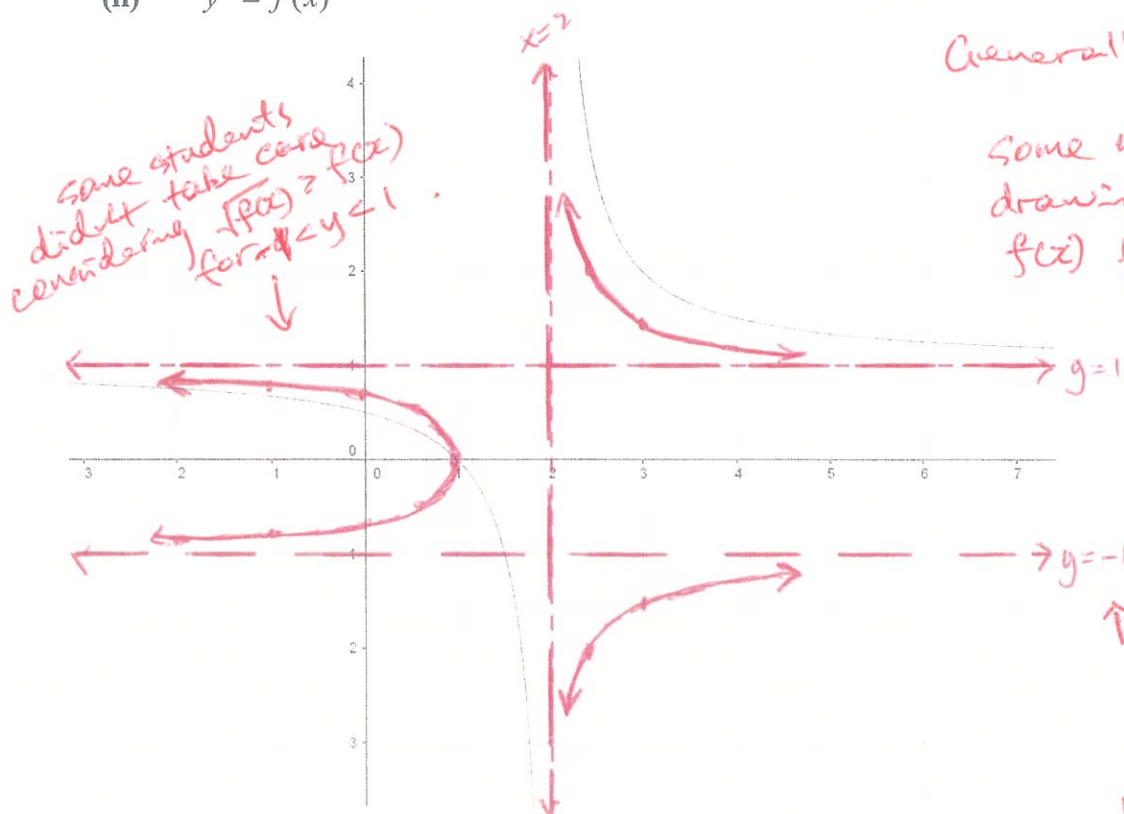
Question 7(a) Place this sheet inside the Question 7 Answer Booklet

(i) $y = |f(x)|$



mostly well done.
Some students
didn't draw on top
of the parts of the
curve included.

(ii) $y^2 = f(x)$



Some students
didn't take care
considering $f(x) > f(x)$
for $x < y < 1$.

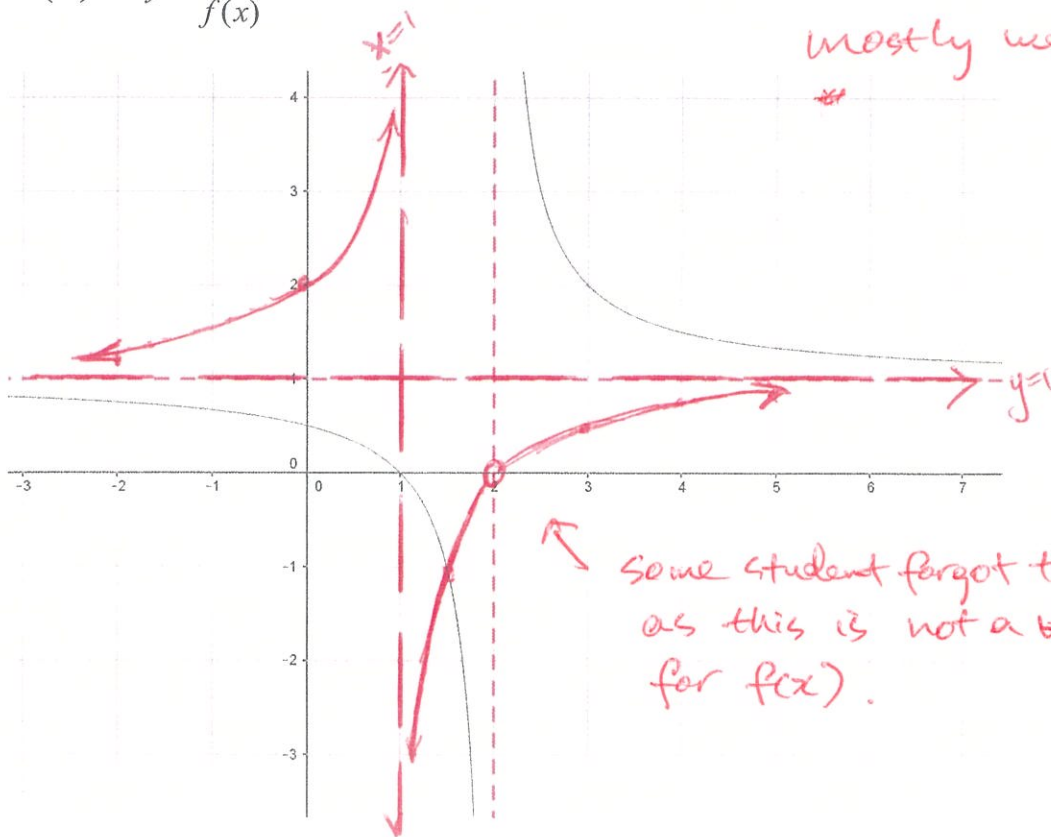
Generally well done.

Some mistakes
drawing $f(x)$ above
 $f(x)$ for $y > 1$.

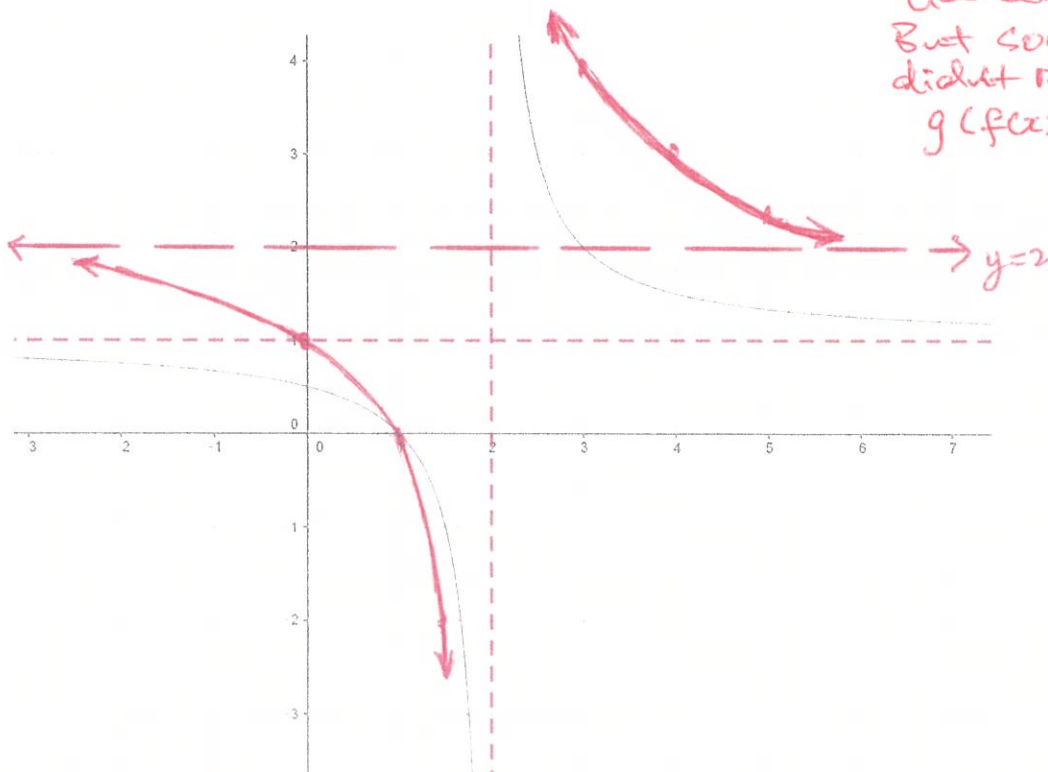
↑ Some students
didn't draw
 $y = -f(x)$ by
reflecting the
curves and
asymptotes.

Question 7(a)

(iii) $y = \frac{1}{f(x)}$



(iv) If $g(x) = 2x$, sketch $y = g[f(x)]$



Generally well done.
But some students
didn't recognise
 $g(f(x)) = 2f(x)$.

many student
forgot to
move the
asymptote up
~~by~~ to $y=2$.